Roll Number:

(8)

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science & Engineering
Auxiliary Test -16th Feb.,2025

BE – 2nd year (Semester-III) **Time: 03 Hours; MM: 100**
UTA018: Object Oriented Programming **In-charge Coordinator: Dr. Palika Chopra**

NOTE: All questions are **mandatory. Pencil work is not permitted.** Follow the given **sequence** and write code **neatly.** Use the **last page for rough work.**

Q1:	#include <iostream>
	using namespace std;
	int main() {
	StringConcatenator obj("Hello, ", "World!");
	obj.concatenateAndDisplay();
	return 0; }
	This assumes you have a StringConcatenator class with a constructor, a member function for concatenation, and a destructor to free allocated memory. Implement class for the same.
		(20)

Q2:	Write short note: a) Constructors b) Polymorphism c) Static variable d) Abstract class	(20)
Q3:	Write a C++ program with two classes, Rectangle and Circle, each with private attributes: • **Rectangle:** length, width • **Circle:** radius Use a **friend function** compareArea(Rectangle r, Circle c) to calculate and compare their areas: Formulas: • **Rectangle Area:** length × width • **Circle Area:** $\pi \times \text{radius}^2$ Take user input for the dimensions and display which object has a larger area or if they are equal.	(20)
Q4:	Write code snippets only: 1. Operator overloading 2. Templates 3. Standard Template Library 4. Inheritance	(20)

Q5:	class Base {
	public:
	Base() { cout << "Base Constructor\n"; }
	~Base() { cout << "Base Destructor\n"; }
	};
	class Derived : public Base {
	public:
	Derived() { cout << "Derived Constructor\n"; }
	~Derived() { cout << "Derived Destructor\n"; }
	};
	int main() {
	Base* ptr = new Derived();
	delete ptr;
	return 0;
	}
	• **Predict the output of the program.**
	• **Identify any issue in the output and explain why it happens.**
	• **How can you modify the code to ensure proper destruction of the derived class object?**
		(20)